

A student needs to separate a mixture of chloroform (bp 61°C) and benzene (bp 80°C). What type(s) of distillation would be expected to give the best separation of the two compounds?

I. Simple distillation

II. Fractional distillation

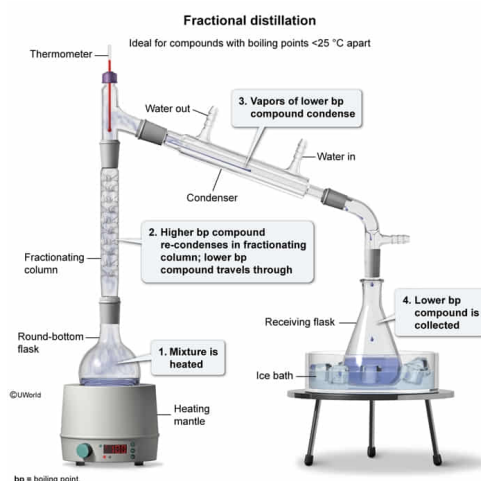
III. Vacuum distillation

- ☐ A. I only [14%]
✓ ☒ B. II only [47%]
☐ C. I and III only [12%]
☐ D. II and III only [24%]

Correct

48% answered correctly

Explanation:



Distillation is a technique used to **separate liquid molecules** based on their **boiling points**. The compound with the lowest boiling point evaporates first. The mixture is heated slowly so that the boiling temperatures of compounds with higher boiling points are not reached before the compound with a lower boiling point (distillate) evaporates and is collected. The vapors of the distillate travel up the distillation column connected to a thermometer and a condenser and then condense into the receiving flask, which is in an ice bath to prevent the distillate from evaporating. This type of distillation is known as a **simple distillation**. It is best suited for compounds with **boiling points <150°C and >25°C apart** from each other. Chloroform and benzene have boiling points separated by <25°C (**Number I**).

There are two variations of simple distillation:

- **Fractional distillation** works best for compounds that have **boiling points** that are **<25°C apart**, such as chloroform and benzene (19°C difference in boiling points). The setup is the same as a simple distillation, with the addition of a fractionating column between the flask and the condenser to provide a larger surface area on which the vapors of the distillate condense before distilling into the receiving flask. The increased area allows for better separation of compounds with similar boiling points (**Number II**).
- **Vacuum distillation** is appropriate for compounds that have **boiling points >150°C**, which may decompose when heated beyond this temperature. To avoid decomposition, the distillation apparatus is connected to a vacuum, decreasing the pressure of the system and consequently lowering the boiling points of the mixture components. Neither chloroform nor benzene has a boiling point >150°C

(Number III).

Educational objective:

Distillation is a purification technique that allows for separation of liquid mixtures based on the boiling points of the mixture's components. The three main types of distillation are simple (bp $<150^{\circ}\text{C}$ and $>25^{\circ}\text{C}$ apart), fractional (bp $<25^{\circ}\text{C}$ apart), and vacuum (bp $>150^{\circ}\text{C}$).

Time spent:0 sec

Subject:
Organic Chemistry

QID:401450

System:
Separation Techniques, Spectroscopy, and Analytical Methods